C4: closed-form time integrals of the second-order radiation-era solutions (Poisson gauge, $a \propto \eta$)

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1 Setup and result in one paragraph

Conventions of SPEC.md and report_C1: $X = \epsilon X_1 + \epsilon^2 X_2/2$, $X_{2,\mathbf{k}} = \int \frac{d^3q}{(2\pi)^3} K_X(q, |\mathbf{k}-\mathbf{q}|; \eta)\, \zeta_{\mathbf{q}} \zeta_{\mathbf{k}-\mathbf{q}}$, $p = |\mathbf{k}-\mathbf{q}|$, $x_a = a\eta/\sqrt{3}$, $T_\Phi(x) = -2(\sin x - x\cos x)/x^3$. The only non-algebraic second-order time integrals are

$$\Phi_2 = P(\eta) + K_{\Phi_2,0}\, \hat{g}(k\eta), \qquad P = \int_0^\eta G_\Phi(\eta,\eta') S_\Phi(\eta') d\eta', \qquad \hat{g} = 3u_1 = -\tfrac{3}{2} T_\Phi(x_k), \tag{1}$$

$$h_{2,ij} = \Lambda_{ij,kl} q_k q_l\, I_h, \qquad I_h = \int_0^\eta \frac{\eta'}{\eta} \frac{\sin k(\eta-\eta')}{k} K_h(\eta') d\eta' = \frac{4}{k^2} I_{\rm RD}^{\rm KT}\left(\tfrac{p}{k}, \tfrac{q}{k}, k\eta\right). \tag{2}$$

Both are obtained in CLOSED FORM: elementary functions plus $\mathrm{Si}(w\eta)$ and the entire function $\mathrm{Cin}(z) = \int_0^z (1-\cos t)\, dt/t = \gamma_E + \ln|z| - \mathrm{Ci}(|z|)$ at the frequencies

$$\Phi_2:\ w_1 = \tfrac{k+q+p}{\sqrt{3}},\ w_2 = \tfrac{k+q-p}{\sqrt{3}},\ w_3 = \tfrac{k-q+p}{\sqrt{3}},\ w_4 = \tfrac{q+p-k}{\sqrt{3}}\ (\text{all } \geq 0); \qquad h_2:\ k \pm \tfrac{q\pm p}{\sqrt{3}}.$$

All other second-order fields are algebraic in Φ_2, Φ_2' and first-order transfer functions (Sec. 5); hence **nothing remains as an unevaluated integral at second order**. Because Cin is entire and even, the forms need no $|\cdot|$, principal values or branch choices; rewriting Cin $\to$ Ci reproduces Kohri–Terada (KT) eq. (22) with its $\log|\ldots|$ terms.

2 Method

The integrands are finite sums $\sum c_{n,w}(k,q,p)\, \eta'^n e^{iw\eta'}$, $-7 \leq n \leq 0$. Each term is integrated by $I_n = [t^{n+1}e^{iwt}]/(n+1) - \frac{iw}{n+1} I_{n+1}$, $I_{-1} = -\mathrm{Cin}(w\eta) + i\,\mathrm{Si}(w\eta) + \ln(\eta/\varepsilon)$, on $[\varepsilon, \eta]$. All ε^{-m} and $\ln\varepsilon$ pieces cancel identically (symbolic check) — this is the regularity of the total integrand at $\eta' = 0$. The source of Φ_2 is

$$\begin{aligned}
S_\Phi = K_{S\Phi} = \Big[\frac{1}{12k^4}\Big(&-7k^6 - 19k^4p^2 - 19k^4q^2 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6 && (3)\\
&+ 7k^2p^4 - 22k^2p^2q^2 + 7k^2q^4\Big)\Big] T_p T_q && (4)\\
+ \Big[&\frac{\eta\left(k^2+q^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}\Big] T_q \dot{T}_p && (5)\\
+ \Big[&\frac{\eta\left(k^2+p^2\right)\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{12k^4}\Big] T_p \dot{T}_q && (6)\\
+ \Big[&\frac{1}{12k^4}\Big(-\eta^2k^6 + 12k^4 - 24k^2p^2 - 24k^2q^2 + 36p^4 - 72p^2q^2 + 36q^4 && (7)\\
&- 2\eta^2k^4p^2 - 2\eta^2k^4q^2 + 3\eta^2k^2p^4 + 3\eta^2k^2q^4 && (8)\\
&- 6\eta^2k^2p^2q^2\Big)\Big] \dot{T}_p \dot{T}_q && (9)
\end{aligned}$$

with $T_a \equiv T_\Phi(x_a)$, $\dot{T}_a \equiv \partial_\eta T_\Phi(x_a)$, and $G_\Phi = [u_1(\eta')u_2(\eta) - u_1(\eta)u_2(\eta')]/W(\eta')$, $u_1 = (\sin x - x\cos x)/x^3$, $u_2 = (\cos x + x\sin x)/x^3$, $W = -3\sqrt{3}/(\eta^4 k^3)$.

3 Structure of Φ_2

$$P = u_2(x_k)\,\mathcal{A}(\eta) - u_1(x_k)\,\mathcal{B}(\eta), \qquad \mathcal{A} = \int_0^\eta \frac{u_1}{W} S_\Phi, \quad \mathcal{B} = \int_0^\eta \frac{u_2}{W} S_\Phi. \tag{10}$$

The special-function skeleton is governed by ONE coefficient,

$$\mathcal{D}(k,q,p) = \frac{(k^2+p^2+q^2)\,R}{8k^2p^3q^3}, \quad R = 3k^6 - 3k^4(p^2+q^2) - 3k^2(p^4+q^4) - 2k^2p^2q^2 + 3(p^6+q^6) - 3p^2q^2(p^2+q^2), \tag{11}$$

$$\mathcal{A} \supset \mathcal{D}\,[\mathrm{Si}(w_1\eta) - \mathrm{Si}(w_2\eta) - \mathrm{Si}(w_3\eta) - \mathrm{Si}(w_4\eta)], \qquad \mathcal{B} \supset -\mathcal{D}\,[\mathrm{Cin}(w_1\eta) - \mathrm{Cin}(w_2\eta) - \mathrm{Cin}(w_3\eta) + \mathrm{Cin}(w_4\eta)], \tag{12}$$

the rest of $\mathcal{A}, \mathcal{B}$ being trigonometric polynomials times Laurent polynomials in η (full form: Appendix A). Only the $n = 0$ monomials produce $1/w$ factors; they become removable $0/0$ at the collinear edges $w_2 = 0$ ($\mu = -1$), $w_3 = 0$ ($\mu = 1, q > k$), $w_4 = 0$ ($\mu = 1, q < k$). Exact collinear limiting forms are obtained by substituting $p = k + q,\ q - k,\ k - q$ in the integrand before integrating (`collinear_closed.pkl`); zero-frequency terms occur only at $n = -4, -2, 0$, so no $\ln\eta$ appears in collinear Φ_2. No $\sqrt{3}$-resonance exists for Φ_2 (sound speed enters both G_Φ and the source).

4 Tensor control: h_2

$I_h = [\sin(k\eta)A_h(\eta) - \cos(k\eta)B_h(\eta)]/\eta$ with $A_h, B_h = \mathrm{Cin/Si}$ at $k \pm (q \pm p)/\sqrt{3}$ plus elementary terms (Appendix B). It equals $(4/k^2)\times$KT(22) [with $|\cdot|$ inside its log] to 10^{-15} (double) / 25 digits (exact inputs). Resonance $q+p = \sqrt{3}k$: harmless in Cin form; the exact resonant form (separately integrated) contains an explicit $\ln\eta$. The other resonance $|q-p| = \sqrt{3}k$ lies outside the physical domain.

5 All other fields (algebraic)

$$\Psi_2 = \Phi_2 - Q_2, \qquad Q_2 = K_Q = \Big[-\frac{13k^4 - 6k^2p^2 - 6k^2q^2 + 9p^4 - 18p^2q^2 + 9q^4}{4k^4} \Big] T_p T_q \tag{13}$$

$$+ \Big[-\frac{\eta\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4} \Big] T_q \dot{T}_p \tag{14}$$

$$+ \Big[-\frac{\eta\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4} \Big] T_p \dot{T}_q \tag{15}$$

$$+ \Big[-\frac{\eta^2\left(-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{4k^4} \Big] \dot{T}_p \dot{T}_q \tag{16}$$

$$\delta_2 = \tfrac{\eta^2}{3} K_{S00} - \tfrac{2}{3}\eta^2 k^2 \Phi_2 - 2\eta\Phi_2' - 2\Psi_2, \qquad \delta_{T2} = \tfrac{1}{4}\delta_2 - \tfrac{3}{16} T_\delta(q) T_\delta(p), \tag{17}$$

$$V_2 = -\tfrac{\eta^2}{4}\Big[2\big(\Phi_2' + \Psi_2/\eta\big) + \frac{K_{S0z}}{ik} \Big], \quad v_{2,i} = \partial_i V_2 + v_{2,i}^T, \tag{18}$$

$$\omega_{2,i} = P_{ij}(\hat{k}) q_j\, K_\omega, \qquad K_\omega = \Big[-\frac{2i\eta\,(p-q)\,(p+q)}{k^2} \Big] T_p T_q \tag{19}$$

$$+ \Big[\frac{2i\eta^2 q^2}{k^2} \Big] T_q \dot{T}_p \tag{20}$$

$$+ \Big[-\frac{2i\eta^2 p^2}{k^2} \Big] T_p \dot{T}_q, \tag{21}$$

$$v_{2,i}^T = P_{ij}(\hat{k}) q_j\, K_{vT} \quad \text{(explicit in second_order.pkl; } K_{vT} = -(\eta^2 k^2/8 + 1) K_\omega - \tfrac{\eta^2}{4} K_{S0\perp}). \tag{22}$$

K_{S00}, K_{S0z} are C1's (closed in T_Φ); Φ_2' is the closed-form derivative ($G_\Phi(\eta,\eta) = 0$).

6 Limits

$\eta \to 0$ **(Frobenius + constraints; checked against the closed forms at 120 digits)**

$$\Phi_2 = \Big[-\frac{7k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{9k^4} \Big] \eta^0 \tag{23}$$

$$+ \Big[\frac{1}{270k^4} \Big(-21k^4p^2 - 21k^4q^2 + 10k^2p^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6 \tag{24}$$

$$- 28k^2p^2q^2 + 10k^2q^4 \Big) \Big] \eta^2 \tag{25}$$

$$+ \Big[\frac{1}{113400k^4} \Big(65k^4p^4 - 80k^2p^6 - 80k^2q^6 - 15p^8 - 36p^6q^2 - 36p^2q^6 - 15q^8 \tag{26}$$

$$+ 150k^6p^2 + 150k^6q^2 + 65k^4q^4 + 144k^2p^2q^4 + 102p^4q^4 \tag{27}$$

$$+ 398k^4p^2q^2 + 144k^2p^4q^2 \Big) \Big] \eta^4 + O(\eta^6) \tag{28}$$

$$I_h = \Big[\frac{8}{9} \Big] \eta^2 \tag{29}$$

$$+ \Big[-\frac{2\left(3k^2 + p^2 + q^2\right)}{135} \Big] \eta^4 + O(\eta^6) \tag{30}$$

$$\Psi_2 = \Big[\frac{2\left(3k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{9k^4} \Big] \eta^0 \tag{31}$$

$$+ \Big[\frac{1}{135k^4} \Big(-16k^4p^2 - 16k^4q^2 + 10k^2p^4 - 6p^6 + 6p^4q^2 + 6p^2q^4 - 6q^6 \tag{32}$$

$$- 4k^2p^2q^2 + 10k^2q^4 \Big) \Big] \eta^2 \tag{33}$$

$$+ \Big[\frac{1}{56700k^4} \Big(55k^4p^4 + 55k^4q^4 - 75k^2p^6 - 75k^2q^6 + 45p^8 - 306p^4q^4 + 45q^8 \tag{34}$$

$$+ 75k^6p^2 + 75k^6q^2 - 117k^2p^2q^4 + 108p^6q^2 + 108p^2q^6 \tag{35}$$

$$+ 234k^4p^2q^2 - 117k^2p^4q^2 \Big) \Big] \eta^4 + O(\eta^6) \tag{36}$$

$$\delta_2 = \Big[-\frac{4\left(-5k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{9k^4} \Big] \eta^0 \tag{37}$$

$$+ \Big[\frac{1}{135k^4} \Big(30k^6 - 82k^4p^2 - 82k^4q^2 - 10k^2p^4 - 10k^2q^4 + 6p^6 + 6q^6 \tag{38}$$

$$+ 4k^2p^2q^2 - 6p^4q^2 - 6p^2q^4 \Big) \Big] \eta^2 \tag{39}$$

$$+ \Big[\frac{1}{28350k^4} \Big(25k^2p^6 + 25k^2q^6 - 15p^8 - 36p^6q^2 + 102p^4q^4 - 36p^2q^6 - 15q^8 \tag{40}$$

$$+ 1095k^6p^2 + 1095k^6q^2 + 275k^4p^4 + 275k^4q^4 + 39k^2p^2q^4 \tag{41}$$

$$+ 3394k^4p^2q^2 + 39k^2p^4q^2 \Big) \Big] \eta^4 + O(\eta^6) \tag{42}$$

$$V_2 = \Big[-\frac{7k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{9k^4} \Big] \eta^1 \tag{43}$$

$$+ \Big[\frac{1}{270k^4} \Big(39k^4p^2 + 6k^2p^4 + 6k^2q^4 + 3p^6 - 3p^4q^2 - 3p^2q^4 + 3q^6 \tag{44}$$

$$+ 39k^4q^2 - 20k^2p^2q^2 \Big) \Big] \eta^3 \tag{45}$$

$$+ \Big[\frac{1}{113400k^4} \Big(-375k^6p^2 - 375k^6q^2 - 105k^2p^6 - 105k^2q^6 - 15p^8 - 15q^8 \tag{46}$$

$$- 105k^4p^4 - 105k^4q^4 - 36p^6q^2 + 102p^4q^4 - 36p^2q^6 \tag{47}$$

$$- 134k^4p^2q^2 + 169k^2p^4q^2 + 169k^2p^2q^4 \Big) \Big] \eta^5 + O(\eta^7) \tag{48}$$

$\eta \to \infty$

With $\Lambda = \ln \frac{(k+q+p)(q+p-k)}{(k+q-p)(k+p-q)}$:

$$\Phi_2 \to \frac{1}{\eta^2}\Big\{C_L\big[\Lambda\cos x_k + \pi \sin x_k\big] + C_0 \cos x_k\Big\} + O(\eta^{-3}), \qquad C_L = -\frac{3(k^2+p^2+q^2)R}{8k^4p^3q^3}, \tag{49}$$

$$C_0 = \frac{3\left[-3k^6 - 3k^4(p^2+q^2) + 3k^2(p^4+q^4) + 2k^2p^2q^2 + 3(p^6+q^6) - 3p^2q^2(p^2+q^2)\right]}{2k^4p^2q^2}, \tag{50}$$

$$I_h \to \frac{4}{k^2}\,\mathrm{KT}(25) \quad (\sin k\eta \text{ log term} + \Theta(q+p-\sqrt{3}k)\,\pi\cos k\eta \text{ term}). \tag{51}$$

Next orders (η^{-3} for Φ_2, η^{-2} for I_h) in `asymptotics.pkl`; verified up to $k\eta = 10^5$ (remainder scales as the first dropped power). The expansion is non-uniform as $q \to 0$ or $p \to 0$ and near the collinear edges / $q + p = \sqrt{3}k$.

7 Verification

- Quadrature arbiter (Gauss–Legendre, 50–60 digits) vs closed forms: P at 11 points incl. collinear and $\sqrt{3}$ points: $\le 3.5 \times 10^{-40}$; I_h: 25 digits; Φ_2, Φ_2', I_h at 10 random points in `verify_module.py`.
- END-TO-END: all C4 fields inserted into C1's full second-order Einstein tensor (10 components) and the 4 conservation equations for a plane-wave pair: residual/max term $\le 5.5\times10^{-50}$ at 3 points ($k\eta = 2.5$, 9.6, 20.1).
- ODE residuals by substitution; small-η series = Frobenius series; KT (22),(25) controls.
- Fast double-precision evaluators (`second_order_closed.py`) vs 50-digit reference: see report.

A Full form of P (particular integral of Φ_2)

$P = u_2(x_k)\mathcal{A} - u_1(x_k)\mathcal{B}$, frequencies $w_1 = (k+q+p)/\sqrt{3}$, $w_2 = (k+q-p)/\sqrt{3}$, $w_3 = (k-q+p)/\sqrt{3}$, $w_4 = (q+p-k)/\sqrt{3}$. Each of $\mathcal{A}, \mathcal{B}$ = special-function part $+ \sum_j [C_j(\eta)\cos w_j\eta + S_j(\eta)\sin w_j\eta] + c_0$.

$\mathcal{A}$

Special functions:

$$\begin{aligned}
&[Si(w_4\eta)]: \ -1\,\mathcal{D}\\
&[Si(w_2\eta)]: \ -1\,\mathcal{D}\\
&[Si(w_3\eta)]: \ -1\,\mathcal{D}\\
&[Si(w_1\eta)]: \ 1\,\mathcal{D}
\end{aligned}$$

constant:

$$c_0 = 0$$

$$\begin{aligned}
C_1 = +\Big[&\frac{27\sqrt{3}\,(k+p+q)}{2k^4p^3q^3}\Big(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\Big)\Big]\eta^{-5}\\
&+\Big[\frac{-3\sqrt{3}}{4k^4p^3q^3}\Big(3k^7 + k^6p + k^6q - 7k^3p^4 - 9k^2p^5 - 9k^2q^5 - 12p^4q^3 - 12p^3q^4\\
&+ 6k^5p^2 + 6k^5q^2 - 7k^3q^4 + 6kp^6 + 6kq^6 + 6p^6q + 6p^2q^5 + 6pq^6\\
&+ 8k^4p^3 + 8k^4q^3 - 13k^2p^4q - 13k^2pq^4 - 36kp^3q^3 + 6p^5q^2\\
&+ 6k^5pq - 12k^3p^3q - 2k^3p^2q^2 - 12k^3pq^3 - 6kp^4q^2 - 6kp^2q^4\\
&+ 10k^4p^2q + 10k^4pq^2 + 14k^2p^3q^2 + 14k^2p^2q^3 + 18kpq^5\\
&+ 18kp^5q\Big)\Big]\eta^{-3}
\end{aligned}$$

$$
\begin{aligned}
&+\Big[\frac{\sqrt{3}}{8k^3p^3q^3}\Big(3k^8-3k^7p-3k^7q-3k^4p^4-3k^2p^6-3k^2q^6-12p^4q^4\\
&-3k^5p^3-3k^5q^3-3k^4q^4+3k^3p^5+3k^3q^5+3kp^7+3kq^7\\
&+3k^6p^2+3k^6q^2-k^2p^4q^2-k^2p^2q^4+6p^6q^2+6p^2q^6\\
&-6k^3p^2q^3-6k^2p^5q-6k^2pq^5-9kp^5q^2-9kp^2q^5\\
&-5k^5p^2q-5k^5pq^2-6k^3p^3q^2+3k^3pq^4-3kp^6q-3kpq^6\\
&+6k^6pq+8k^4p^3q+4k^4p^2q^2+8k^4pq^3+3k^3p^4q+9kp^3q^4\\
&+12k^2p^3q^3+9kp^4q^3\Big)\Big]\eta^{-1}\\
S_1=&+\Big[-\frac{81\left(k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6}\\
&+\Big[\frac{1}{4k^4p^3q^3}\Big(9k^6-108k^3p^3-117k^2p^4-117k^2q^4+54p^6-324p^3q^3+54q^6\\
&+54k^5p+54k^5q+90k^4p^2+90k^4q^2-108k^3q^3-54p^4q^2-54p^2q^4\\
&+162kp^5-324kp^3q^2-324kp^2q^3+162kq^5+162p^5q+162pq^5\\
&+54k^4pq-108k^3p^2q-108k^3pq^2-108k^2p^3q-108k^2pq^3\\
&+90k^2p^2q^2+162kp^4q+162kpq^4\Big)\Big]\eta^{-4}\\
&+\Big[\frac{1}{8k^4p^3q^3}\Big(9k^8-18k^7p-18k^7q-24k^5p^3-24k^5q^3-18p^6q^2-18p^2q^6\\
&+15k^6p^2+15k^6q^2-9k^4p^4-9k^4q^4+9k^2p^6+9k^2q^6\\
&-48k^4pq^3+18k^3p^5+18k^3q^5-108k^2p^3q^3+36p^4q^4\\
&-36k^5pq^2-48k^4p^3q-36kp^6q-36kp^5q^2-36kp^2q^5-36kpq^6\\
&-6k^6pq-36k^5p^2q-12k^3p^3q^2-12k^3p^2q^3+3k^2p^2q^4\\
&+24k^4p^2q^2+42k^3p^4q+42k^3pq^4+3k^2p^4q^2+54k^2pq^5\\
&+54k^2p^5q+72kp^4q^3+72kp^3q^4\Big)\Big]\eta^{-2}\\
&+\Big[-\frac{-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4}{4kpq\left(k+p+q\right)}\Big]\eta^{0}\\
C_2=&+\Big[\frac{27\sqrt{3}\left(-k+p-q\right)}{2k^4p^3q^3}\Big(k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\Big)\Big]\eta^{-5}\\
&+\Big[\frac{3\sqrt{3}}{4k^4p^3q^3}\Big(3k^7-k^6p+k^6q-8k^4p^3-7k^3p^4-7k^3q^4-9k^2q^5-12p^4q^3\\
&+6k^5p^2+6k^5q^2+6kp^6+6kq^6+6p^6q-6p^5q^2+6p^2q^5-6pq^6\\
&+8k^4q^3+9k^2p^5-14k^2p^3q^2-18kp^5q-18kpq^5+12p^3q^4\\
&-6k^5pq-10k^4pq^2-2k^3p^2q^2-13k^2p^4q-6kp^4q^2-6kp^2q^4\\
&+10k^4p^2q+12k^3p^3q+12k^3pq^3+14k^2p^2q^3+13k^2pq^4\\
&+36kp^3q^3\Big)\Big]\eta^{-3}\\
&+\Big[\frac{\sqrt{3}}{8k^3p^3q^3}\Big(-3k^8-3k^7p-3k^5p^3-3k^3q^5-3kq^7-6p^6q^2-6p^2q^6\\
&+3k^7q-3k^6p^2-3k^6q^2+3k^3p^5+3k^2p^6+3k^2q^6+3kp^7\\
&+3k^5q^3+3k^4p^4+3k^4q^4+k^2p^4q^2+k^2p^2q^4+12p^4q^4\\
&-6k^3p^3q^2-6k^2p^5q-6k^2pq^5-9kp^5q^2-9kp^4q^3\\
&-5k^5pq^2-4k^4p^2q^2-3k^3p^4q+3k^3pq^4+3kp^6q-3kpq^6\\
&+6k^6pq+5k^5p^2q+8k^4p^3q+8k^4pq^3+6k^3p^2q^3+9kp^2q^5
\end{aligned}
$$

$$
\begin{aligned}
&+ 12k^2p^3q^3 + 9kp^3q^4\Big)\Big]\eta^{-1}\\
S_2 = &+\Big[\frac{81\left(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6}\\
&+\Big[\frac{1}{4k^4p^3q^3}\Big(-9k^6 - 90k^4p^2 - 108k^3p^3 - 162kq^5 - 54p^6 - 324p^3q^3 - 54q^6\\
&+ 54k^5p - 54k^5q - 90k^4q^2 + 108k^3q^3 + 117k^2p^4 + 54p^4q^2 + 54p^2q^4\\
&+ 117k^2q^4 + 162kp^5 - 162kp^4q - 324kp^3q^2 + 162p^5q + 162pq^5\\
&+ 54k^4pq - 108k^3pq^2 - 108k^2p^3q - 90k^2p^2q^2 - 108k^2pq^3\\
&+ 108k^3p^2q + 324kp^2q^3 + 162kpq^4\Big)\Big]\eta^{-4}\\
&+\Big[\frac{1}{8k^4p^3q^3}\Big(-9k^8 - 18k^7p - 15k^6p^2 - 15k^6q^2 - 24k^5p^3 - 18k^3q^5 - 36p^4q^4\\
&+ 18k^7q + 9k^4p^4 + 9k^4q^4 + 18k^3p^5 - 9k^2p^6 - 9k^2q^6 + 18p^2q^6\\
&+ 24k^5q^3 - 48k^4pq^3 - 108k^2p^3q^3 - 72kp^4q^3 + 18p^6q^2\\
&- 36k^5pq^2 - 48k^4p^3q - 42k^3p^4q - 36kp^5q^2 - 36kpq^6\\
&- 6k^6pq - 24k^4p^2q^2 - 12k^3p^3q^2 - 3k^2p^4q^2 - 3k^2p^2q^4\\
&+ 36k^5p^2q + 12k^3p^2q^3 + 42k^3pq^4 + 36kp^6q + 36kp^2q^5\\
&+ 54k^2p^5q + 54k^2pq^5 + 72kp^3q^4\Big)\Big]\eta^{-2}\\
&+\Big[-\frac{-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{4kpq\left(-k + p - q\right)}\Big]\eta^{0}\\
C_3 = &+\Big[\frac{-27\sqrt{3}\left(k + p - q\right)}{2k^4p^3q^3}\Big(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\Big)\Big]\eta^{-5}\\
&+\Big[\frac{-3\sqrt{3}}{4k^4p^3q^3}\Big(-3k^7 - k^6p + k^6q - 8k^4p^3 - 9k^2q^5 - 6kp^6 - 6kq^6 - 12p^4q^3\\
&- 6k^5p^2 - 6k^5q^2 + 7k^3p^4 + 6p^6q - 6p^5q^2 + 6p^2q^5 - 6pq^6\\
&+ 8k^4q^3 + 7k^3q^4 + 9k^2p^5 - 14k^2p^3q^2 - 36kp^3q^3 + 12p^3q^4\\
&- 10k^4pq^2 - 12k^3p^3q + 2k^3p^2q^2 - 12k^3pq^3 - 13k^2p^4q\\
&+ 6k^5pq + 10k^4p^2q + 14k^2p^2q^3 + 13k^2pq^4 + 6kp^4q^2 + 6kp^2q^4\\
&+ 18kp^5q + 18kpq^5\Big)\Big]\eta^{-3}\\
&+\Big[\frac{-\sqrt{3}}{8k^3p^3q^3}\Big(3k^8 - 3k^7p - 3k^3q^5 - 3k^2p^6 - 3k^2q^6 - 3kq^7 - 12p^4q^4\\
&+ 3k^7q - 3k^5p^3 + 3k^5q^3 - 3k^4p^4 - 3k^4q^4 + 3k^3p^5 + 3kp^7\\
&+ 3k^6p^2 + 3k^6q^2 - k^2p^4q^2 - k^2p^2q^4 + 6p^6q^2 + 6p^2q^6\\
&- 6k^6pq - 8k^4p^3q - 8k^4pq^3 - 12k^2p^3q^3 - 9kp^5q^2 - 9kp^4q^3\\
&- 5k^5pq^2 - 3k^3p^4q - 6k^3p^3q^2 + 3k^3pq^4 + 3kp^6q - 3kpq^6\\
&+ 5k^5p^2q + 4k^4p^2q^2 + 6k^3p^2q^3 + 6k^2p^5q + 6k^2pq^5\\
&+ 9kp^3q^4 + 9kp^2q^5\Big)\Big]\eta^{-1}\\
S_3 = &+\Big[\frac{81\left(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6}\\
&+\Big[\frac{1}{4k^4p^3q^3}\Big(-9k^6 - 90k^4p^2 - 108k^3q^3 - 162kp^5 - 54p^6 - 324p^3q^3 - 54q^6\\
&- 54k^5p + 54k^5q - 90k^4q^2 + 108k^3p^3 + 54p^4q^2 + 54p^2q^4\\
&+ 117k^2p^4 + 117k^2q^4 - 324kp^2q^3 + 162kq^5 + 162p^5q + 162pq^5\\
&- 108k^3p^2q - 108k^2p^3q - 90k^2p^2q^2 - 108k^2pq^3 - 162kpq^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 54k^4pq + 108k^3pq^2 + 162kp^4q + 324kp^3q^2\Big)\Big]\eta^{-4} \\
&+ \Big[\frac{1}{8k^4p^3q^3}\Big(- 9k^8 - 18k^7q - 15k^6p^2 - 15k^6q^2 - 24k^5q^3 - 18k^3p^5 - 36p^4q^4 \\
&+ 18k^7p + 9k^4p^4 + 9k^4q^4 + 18k^3q^5 - 9k^2p^6 - 9k^2q^6 + 18p^2q^6 \\
&+ 24k^5p^3 - 48k^4pq^3 - 108k^2p^3q^3 - 72kp^3q^4 + 18p^6q^2 \\
&- 36k^5p^2q - 48k^4p^3q - 42k^3pq^4 - 36kp^6q - 36kp^2q^5 \\
&- 6k^6pq - 24k^4p^2q^2 - 12k^3p^2q^3 - 3k^2p^4q^2 - 3k^2p^2q^4 \\
&+ 36k^5pq^2 + 42k^3p^4q + 12k^3p^3q^2 + 36kp^5q^2 + 36kpq^6 \\
&+ 54k^2p^5q + 54k^2pq^5 + 72kp^4q^3\Big)\Big]\eta^{-2} \\
&+ \Big[\frac{-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{4kpq\,(k+p-q)}\Big]\eta^{0} \\
C_4 = &+ \Big[\frac{-27\sqrt{3}\,(-k+p+q)}{2k^4p^3q^3}\Big(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\Big)\Big]\eta^{-5} \\
&+ \Big[\frac{3\sqrt{3}}{4k^4p^3q^3}\Big(- 3k^7 + k^6p + k^6q - 9k^2p^5 - 9k^2q^5 - 6kp^6 - 12p^4q^3 - 12p^3q^4 \\
&- 6k^5p^2 - 6k^5q^2 - 6kq^6 + 6p^6q + 6p^5q^2 + 6p^2q^5 + 6pq^6 \\
&+ 8k^4p^3 + 8k^4q^3 + 7k^3p^4 + 7k^3q^4 - 13k^2pq^4 - 18kp^5q - 18kpq^5 \\
&- 6k^5pq + 10k^4pq^2 + 2k^3p^2q^2 - 13k^2p^4q + 6kp^4q^2 + 6kp^2q^4 \\
&+ 10k^4p^2q + 12k^3p^3q + 12k^3pq^3 + 14k^2p^3q^2 + 14k^2p^2q^3 \\
&+ 36kp^3q^3\Big)\Big]\eta^{-3} \\
&+ \Big[\frac{-\sqrt{3}}{8k^3p^3q^3}\Big(- 3k^8 - 3k^7p - 3k^7q - 3k^5p^3 - 3k^5q^3 - 6p^6q^2 - 6p^2q^6 \\
&- 3k^6p^2 - 3k^6q^2 + 3k^3p^5 + 3k^2p^6 + 3k^2q^6 + 3kp^7 + 3kq^7 \\
&+ 3k^4p^4 + 3k^4q^4 + 3k^3q^5 + k^2p^4q^2 + k^2p^2q^4 + 12p^4q^4 \\
&- 6k^6pq - 8k^4p^3q - 8k^4pq^3 - 12k^2p^3q^3 - 9kp^5q^2 - 9kp^2q^5 \\
&- 5k^5p^2q - 5k^5pq^2 - 4k^4p^2q^2 - 6k^3p^3q^2 - 6k^3p^2q^3 \\
&+ 3k^3p^4q + 3k^3pq^4 + 6k^2p^5q + 6k^2pq^5 - 3kp^6q - 3kpq^6 \\
&+ 9kp^4q^3 + 9kp^3q^4\Big)\Big]\eta^{-1} \\
S_4 = &+ \Big[\frac{81\left(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6} \\
&+ \Big[\frac{1}{4k^4p^3q^3}\Big(- 9k^6 - 108k^3p^3 - 108k^3q^3 - 54p^6 - 162p^5q - 162pq^5 - 54q^6 \\
&+ 54k^5p + 54k^5q - 90k^4p^2 - 90k^4q^2 + 117k^2p^4 + 54p^4q^2 + 54p^2q^4 \\
&+ 117k^2q^4 + 162kp^5 - 324kp^3q^2 - 324kp^2q^3 + 162kq^5 + 324p^3q^3 \\
&- 54k^4pq - 108k^3p^2q - 108k^3pq^2 - 90k^2p^2q^2 + 108k^2pq^3 \\
&+ 108k^2p^3q + 162kp^4q + 162kpq^4\Big)\Big]\eta^{-4} \\
&+ \Big[\frac{1}{8k^4p^3q^3}\Big(- 9k^8 - 18k^7p - 18k^7q - 15k^6p^2 - 24k^5p^3 - 24k^5q^3 - 36p^4q^4 \\
&- 15k^6q^2 + 9k^4p^4 + 9k^4q^4 + 18k^3p^5 - 9k^2p^6 - 9k^2q^6 \\
&+ 18k^3q^5 - 54k^2p^5q - 54k^2pq^5 - 36kpq^6 + 18p^6q^2 + 18p^2q^6 \\
&- 36k^5p^2q - 36k^5pq^2 - 36kp^6q - 36kp^5q^2 - 36kp^2q^5 \\
&- 24k^4p^2q^2 - 12k^3p^3q^2 - 12k^3p^2q^3 - 3k^2p^2q^4
\end{aligned}
$$

$$
\begin{aligned}
&+ 6k^6pq + 48k^4p^3q + 48k^4pq^3 + 42k^3p^4q + 42k^3pq^4 - 3k^2p^4q^2 \\
&+ 108k^2p^3q^3 + 72kp^4q^3 + 72kp^3q^4\Big)\Big]\eta^{-2} \\
&+ \Big[- \frac{-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{4kpq\,(-k+p+q)}\Big]\eta^{0}
\end{aligned}
$$

$\mathcal{B}$

Special functions:

$$
\begin{aligned}
&[Cin(w_1\eta)]: \ -1\,\mathcal{D} \\
&[Cin(w_3\eta)]: \ 1\,\mathcal{D} \\
&[Cin(w_2\eta)]: \ 1\,\mathcal{D} \\
&[Cin(w_4\eta)]: \ -1\,\mathcal{D}
\end{aligned}
$$

constant:

$$
\begin{aligned}
c_0 = \frac{1}{6k^4p^2q^2\,(-k+p-q)\,(-k+p+q)\,(k+p-q)\,(k+p+q)}\Big(& - 9k^{12} - 18k^6p^6 - 18k^6q^6 - 9k^4p^8 - 9k^4q^8 - 36p^6q^6 - 6p^2q^{10} \\
& + 9k^{10}p^2 + 9k^{10}q^2 + 18k^8p^4 + 9k^2p^{10} + 9k^2q^{10} - 6p^{10}q^2 \\
& + 18k^8q^4 - 22k^6p^4q^2 - 22k^6p^2q^4 + 24p^8q^4 + 24p^4q^8 \\
& - 6k^4p^4q^4 - 4k^4p^2q^6 - 11k^2p^8q^2 - 11k^2p^2q^8 \\
& + 34k^8p^2q^2 - 4k^4p^6q^2 + 2k^2p^6q^4 + 2k^2p^4q^6\Big)
\end{aligned}
$$

$$
\begin{aligned}
C_1 = +\Big[& - \frac{81\left(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6} \\
& + \Big[\frac{1}{4k^4p^3q^3}\Big(9k^6 - 108k^3p^3 - 117k^2p^4 - 117k^2q^4 + 54p^6 - 324p^3q^3 + 54q^6 \\
& + 54k^5p + 54k^5q + 90k^4p^2 + 90k^4q^2 - 108k^3q^3 - 54p^4q^2 - 54p^2q^4 \\
& + 162kp^5 - 324kp^3q^2 - 324kp^2q^3 + 162kq^5 + 162p^5q + 162pq^5 \\
& + 54k^4pq - 108k^3p^2q - 108k^3pq^2 - 108k^2p^3q - 108k^2pq^3 \\
& + 90k^2p^2q^2 + 162kp^4q + 162kpq^4\Big)\Big]\eta^{-4} \\
& + \Big[\frac{1}{8k^4p^3q^3}\Big(9k^8 - 18k^7p - 18k^7q - 24k^5p^3 - 24k^5q^3 - 18p^6q^2 - 18p^2q^6 \\
& + 15k^6p^2 + 15k^6q^2 - 9k^4p^4 - 9k^4q^4 + 9k^2p^6 + 9k^2q^6 \\
& - 48k^4pq^3 + 18k^3p^5 + 18k^3q^5 - 108k^2p^3q^3 + 36p^4q^4 \\
& - 36k^5pq^2 - 48k^4p^3q - 36kp^6q - 36kp^5q^2 - 36kp^2q^5 - 36kpq^6 \\
& - 6k^6pq - 36k^5p^2q - 12k^3p^3q^2 - 12k^3p^2q^3 + 3k^2p^2q^4 \\
& + 24k^4p^2q^2 + 42k^3p^4q + 42k^3pq^4 + 3k^2p^4q^2 + 54k^2pq^5 \\
& + 54k^2p^5q + 72kp^4q^3 + 72kp^3q^4\Big)\Big]\eta^{-2} \\
& + \Big[- \frac{-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{4kpq\,(k+p+q)}\Big]\eta^{0}
\end{aligned}
$$

$$
\begin{aligned}
S_1 = +\Big[& \frac{-27\sqrt{3}\,(k+p+q)}{2k^4p^3q^3}\Big(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\Big)\Big]\eta^{-5} \\
& + \Big[\frac{3\sqrt{3}}{4k^4p^3q^3}\Big(3k^7 + k^6p + k^6q - 7k^3p^4 - 9k^2p^5 - 9k^2q^5 - 12p^4q^3 - 12p^3q^4 \\
& + 6k^5p^2 + 6k^5q^2 - 7k^3q^4 + 6kp^6 + 6kq^6 + 6p^6q + 6p^2q^5 + 6pq^6
\end{aligned}
$$

$$
\begin{aligned}
&+ 8k^4p^3 + 8k^4q^3 - 13k^2p^4q - 13k^2pq^4 - 36kp^3q^3 + 6p^5q^2 \\
&+ 6k^5pq - 12k^3p^3q - 2k^3p^2q^2 - 12k^3pq^3 - 6kp^4q^2 - 6kp^2q^4 \\
&+ 10k^4p^2q + 10k^4pq^2 + 14k^2p^3q^2 + 14k^2p^2q^3 + 18kpq^5 \\
&+ 18kp^5q\Big)\Big]\eta^{-3} \\
&+ \Big[\frac{-\sqrt{3}}{8k^3p^3q^3}\Big(3k^8 - 3k^7p - 3k^7q - 3k^4p^4 - 3k^2p^6 - 3k^2q^6 - 12p^4q^4 \\
&- 3k^5p^3 - 3k^5q^3 - 3k^4q^4 + 3k^3p^5 + 3k^3q^5 + 3kp^7 + 3kq^7 \\
&+ 3k^6p^2 + 3k^6q^2 - k^2p^4q^2 - k^2p^2q^4 + 6p^6q^2 + 6p^2q^6 \\
&- 6k^3p^2q^3 - 6k^2p^5q - 6k^2pq^5 - 9kp^5q^2 - 9kp^2q^5 \\
&- 5k^5p^2q - 5k^5pq^2 - 6k^3p^3q^2 + 3k^3pq^4 - 3kp^6q - 3kpq^6 \\
&+ 6k^6pq + 8k^4p^3q + 4k^4p^2q^2 + 8k^4pq^3 + 3k^3p^4q + 9kp^3q^4 \\
&+ 12k^2p^3q^3 + 9kp^4q^3\Big)\Big]\eta^{-1} \\
C_2 = &+\Big[\frac{81\left(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6} \\
&+ \Big[\frac{1}{4k^4p^3q^3}\Big(-9k^6 - 90k^4p^2 - 108k^3p^3 - 162kq^5 - 54p^6 - 324p^3q^3 - 54q^6 \\
&+ 54k^5p - 54k^5q - 90k^4q^2 + 108k^3q^3 + 117k^2p^4 + 54p^4q^2 + 54p^2q^4 \\
&+ 117k^2q^4 + 162kp^5 - 162kp^4q - 324kp^3q^2 + 162p^5q + 162pq^5 \\
&+ 54k^4pq - 108k^3pq^2 - 108k^2p^3q - 90k^2p^2q^2 - 108k^2pq^3 \\
&+ 108k^3p^2q + 324kp^2q^3 + 162kpq^4\Big)\Big]\eta^{-4} \\
&+ \Big[\frac{1}{8k^4p^3q^3}\Big(-9k^8 - 18k^7p - 15k^6p^2 - 15k^6q^2 - 24k^5p^3 - 18k^3q^5 - 36p^4q^4 \\
&+ 18k^7q + 9k^4p^4 + 9k^4q^4 + 18k^3p^5 - 9k^2p^6 - 9k^2q^6 + 18p^2q^6 \\
&+ 24k^5q^3 - 48k^4pq^3 - 108k^2p^3q^3 - 72kp^4q^3 + 18p^6q^2 \\
&- 36k^5pq^2 - 48k^4p^3q - 42k^3p^4q - 36kp^5q^2 - 36kpq^6 \\
&- 6k^6pq - 24k^4p^2q^2 - 12k^3p^3q^2 - 3k^2p^4q^2 - 3k^2p^2q^4 \\
&+ 36k^5p^2q + 12k^3p^2q^3 + 42k^3pq^4 + 36kp^6q + 36kp^2q^5 \\
&+ 54k^2p^5q + 54k^2pq^5 + 72kp^3q^4\Big)\Big]\eta^{-2} \\
&+ \Big[-\frac{-k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4}{4kpq\left(-k + p - q\right)}\Big]\eta^{0} \\
S_2 = &+\Big[\frac{-27\sqrt{3}\left(-k + p - q\right)}{2k^4p^3q^3}\Big(k^4 - 2k^2p^2 - 2k^2q^2 + 3p^4 - 6p^2q^2 + 3q^4\Big)\Big]\eta^{-5} \\
&+ \Big[\frac{-3\sqrt{3}}{4k^4p^3q^3}\Big(3k^7 - k^6p + k^6q - 8k^4p^3 - 7k^3p^4 - 7k^3q^4 - 9k^2q^5 - 12p^4q^3 \\
&+ 6k^5p^2 + 6k^5q^2 + 6kp^6 + 6kq^6 + 6p^6q - 6p^5q^2 + 6p^2q^5 - 6pq^6 \\
&+ 8k^4q^3 + 9k^2p^5 - 14k^2p^3q^2 - 18kp^5q - 18kpq^5 + 12p^3q^4 \\
&- 6k^5pq - 10k^4pq^2 - 2k^3p^2q^2 - 13k^2p^4q - 6kp^4q^2 - 6kp^2q^4 \\
&+ 10k^4p^2q + 12k^3p^3q + 12k^3pq^3 + 14k^2p^2q^3 + 13k^2pq^4 \\
&+ 36kp^3q^3\Big)\Big]\eta^{-3} \\
&+ \Big[\frac{-\sqrt{3}}{8k^3p^3q^3}\Big(-3k^8 - 3k^7p - 3k^5p^3 - 3k^3q^5 - 3kq^7 - 6p^6q^2 - 6p^2q^6 \\
&+ 3k^7q - 3k^6p^2 - 3k^6q^2 + 3k^3p^5 + 3k^2p^6 + 3k^2q^6 + 3kp^7
\end{aligned}
$$

$$
\begin{aligned}
&+3k^5q^3+3k^4p^4+3k^4q^4+k^2p^4q^2+k^2p^2q^4+12p^4q^4\\
&-6k^3p^3q^2-6k^2p^5q-6k^2pq^5-9kp^5q^2-9kp^4q^3\\
&-5k^5pq^2-4k^4p^2q^2-3k^3p^4q+3k^3pq^4+3kp^6q-3kpq^6\\
&+6k^6pq+5k^5p^2q+8k^4p^3q+8k^4pq^3+6k^3p^2q^3+9kp^2q^5\\
&+12k^2p^3q^3+9kp^3q^4\Big)\Big]\eta^{-1}\\
C_3=+\Big[&\frac{81\left(k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6}\\
&+\Big[\frac{1}{4k^4p^3q^3}\Big(-9k^6-90k^4p^2-108k^3q^3-162kp^5-54p^6-324p^3q^3-54q^6\\
&-54k^5p+54k^5q-90k^4q^2+108k^3p^3+54p^4q^2+54p^2q^4\\
&+117k^2p^4+117k^2q^4-324kp^2q^3+162kq^5+162p^5q+162pq^5\\
&-108k^3p^2q-108k^2p^3q-90k^2p^2q^2-108k^2pq^3-162kpq^4\\
&+54k^4pq+108k^3pq^2+162kp^4q+324kp^3q^2\Big)\Big]\eta^{-4}\\
&+\Big[\frac{1}{8k^4p^3q^3}\Big(-9k^8-18k^7q-15k^6p^2-15k^6q^2-24k^5q^3-18k^3p^5-36p^4q^4\\
&+18k^7p+9k^4p^4+9k^4q^4+18k^3q^5-9k^2p^6-9k^2q^6+18p^2q^6\\
&+24k^5p^3-48k^4pq^3-108k^2p^3q^3-72kp^3q^4+18p^6q^2\\
&-36k^5p^2q-48k^4p^3q-42k^3pq^4-36kp^6q-36kp^2q^5\\
&-6k^6pq-24k^4p^2q^2-12k^3p^2q^3-3k^2p^4q^2-3k^2p^2q^4\\
&+36k^5pq^2+42k^3p^4q+12k^3p^3q^2+36kp^5q^2+36kpq^6\\
&+54k^2p^5q+54k^2pq^5+72kp^4q^3\Big)\Big]\eta^{-2}\\
&+\Big[\frac{-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4}{4kpq\left(k+p-q\right)}\Big]\eta^{0}\\
S_3=+\Big[&\frac{27\sqrt{3}\left(k+p-q\right)}{2k^4p^3q^3}\Big(k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\Big)\Big]\eta^{-5}\\
&+\Big[\frac{3\sqrt{3}}{4k^4p^3q^3}\Big(-3k^7-k^6p+k^6q-8k^4p^3-9k^2q^5-6kp^6-6kq^6-12p^4q^3\\
&-6k^5p^2-6k^5q^2+7k^3p^4+6p^6q-6p^5q^2+6p^2q^5-6pq^6\\
&+8k^4q^3+7k^3q^4+9k^2p^5-14k^2p^3q^2-36kp^3q^3+12p^3q^4\\
&-10k^4pq^2-12k^3p^3q+2k^3p^2q^2-12k^3pq^3-13k^2p^4q\\
&+6k^5pq+10k^4p^2q+14k^2p^2q^3+13k^2pq^4+6kp^4q^2+6kp^2q^4\\
&+18kp^5q+18kpq^5\Big)\Big]\eta^{-3}\\
&+\Big[\frac{\sqrt{3}}{8k^3p^3q^3}\Big(3k^8-3k^7p-3k^3q^5-3k^2p^6-3k^2q^6-3kq^7-12p^4q^4\\
&+3k^7q-3k^5p^3+3k^5q^3-3k^4p^4-3k^4q^4+3k^3p^5+3kp^7\\
&+3k^6p^2+3k^6q^2-k^2p^4q^2-k^2p^2q^4+6p^6q^2+6p^2q^6\\
&-6k^6pq-8k^4p^3q-8k^4pq^3-12k^2p^3q^3-9kp^5q^2-9kp^4q^3\\
&-5k^5pq^2-3k^3p^4q-6k^3p^3q^2+3k^3pq^4+3kp^6q-3kpq^6\\
&+5k^5p^2q+4k^4p^2q^2+6k^3p^2q^3+6k^2p^5q+6k^2pq^5\\
&+9kp^3q^4+9kp^2q^5\Big)\Big]\eta^{-1}\\
C_4=+\Big[&-\frac{81\left(k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\right)}{2k^4p^3q^3}\Big]\eta^{-6}
\end{aligned}
$$

$$
\begin{aligned}
&+\Big[\frac{1}{4k^4p^3q^3}\Big(9k^6-117k^2p^4-162kp^5-162kq^5+54p^6-324p^3q^3+54q^6\\
&-54k^5p-54k^5q+90k^4p^2-117k^2q^4-54p^4q^2-54p^2q^4\\
&+90k^4q^2+108k^3p^3+108k^3q^3-162kp^4q-162kpq^4+162p^5q+162pq^5\\
&+54k^4pq+108k^3pq^2-108k^2p^3q+90k^2p^2q^2-108k^2pq^3\\
&+108k^3p^2q+324kp^3q^2+324kp^2q^3\Big)\Big]\eta^{-4}\\
&+\Big[\frac{1}{8k^4p^3q^3}\Big(9k^8-9k^4p^4-9k^4q^4-18k^3p^5-18k^3q^5-18p^6q^2-18p^2q^6\\
&+18k^7p+18k^7q+15k^6p^2+15k^6q^2+24k^5p^3+9k^2p^6+9k^2q^6\\
&+24k^5q^3-108k^2p^3q^3-72kp^4q^3-72kp^3q^4+36p^4q^4\\
&-6k^6pq-48k^4p^3q-48k^4pq^3-42k^3p^4q-42k^3pq^4\\
&+24k^4p^2q^2+12k^3p^3q^2+12k^3p^2q^3+3k^2p^4q^2+3k^2p^2q^4\\
&+36k^5p^2q+36k^5pq^2+36kp^6q+36kp^5q^2+36kp^2q^5+36kpq^6\\
&+54k^2p^5q+54k^2pq^5\Big)\Big]\eta^{-2}\\
&+\Big[\frac{-k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4}{4kpq\,(-k+p+q)}\Big]\eta^{0}\\
S_4=&+\Big[\frac{-27\sqrt{3}\,(-k+p+q)}{2k^4p^3q^3}\Big(k^4-2k^2p^2-2k^2q^2+3p^4-6p^2q^2+3q^4\Big)\Big]\eta^{-5}\\
&+\Big[\frac{3\sqrt{3}}{4k^4p^3q^3}\Big(-3k^7+k^6p+k^6q-9k^2p^5-9k^2q^5-6kp^6-12p^4q^3-12p^3q^4\\
&-6k^5p^2-6k^5q^2-6kq^6+6p^6q+6p^5q^2+6p^2q^5+6pq^6\\
&+8k^4p^3+8k^4q^3+7k^3p^4+7k^3q^4-13k^2pq^4-18kp^5q-18kpq^5\\
&-6k^5pq+10k^4pq^2+2k^3p^2q^2-13k^2p^4q+6kp^4q^2+6kp^2q^4\\
&+10k^4p^2q+12k^3p^3q+12k^3pq^3+14k^2p^3q^2+14k^2p^2q^3\\
&+36kp^3q^3\Big)\Big]\eta^{-3}\\
&+\Big[\frac{-\sqrt{3}}{8k^3p^3q^3}\Big(-3k^8-3k^7p-3k^7q-3k^5p^3-3k^5q^3-6p^6q^2-6p^2q^6\\
&-3k^6p^2-3k^6q^2+3k^3p^5+3k^2p^6+3k^2q^6+3kp^7+3kq^7\\
&+3k^4p^4+3k^4q^4+3k^3q^5+k^2p^4q^2+k^2p^2q^4+12p^4q^4\\
&-6k^6pq-8k^4p^3q-8k^4pq^3-12k^2p^3q^3-9kp^5q^2-9kp^2q^5\\
&-5k^5p^2q-5k^5pq^2-4k^4p^2q^2-6k^3p^3q^2-6k^3p^2q^3\\
&+3k^3p^4q+3k^3pq^4+6k^2p^5q+6k^2pq^5-3kp^6q-3kpq^6\\
&+9kp^4q^3+9kp^3q^4\Big)\Big]\eta^{-1}
\end{aligned}
$$

B Full form of I_h

Frequencies: $w_1=k+(q+p)/\sqrt{3},\ w_2=k+(q-p)/\sqrt{3},\ w_3=k-(q-p)/\sqrt{3},\ w_4=k-(q+p)/\sqrt{3}$ (only w_4 can vanish). Special-function part (function, frequency):

$$
[Cin,\ w_2]:\ -\frac{3\left(-3k^2+p^2+q^2\right)^2\sin(\eta k)}{\eta k p^3q^3}
$$

$$
[Cin,\ w_3]:\ -\frac{3\left(-3k^2+p^2+q^2\right)^2\sin(\eta k)}{\eta k p^3q^3}
$$

$$[Cin,\ w_1]:\ \frac{3\left(-3k^2+p^2+q^2\right)^2\sin\left(\eta k\right)}{\eta k p^3 q^3}$$

$$[Cin,\ w_4]:\ \frac{3\left(-3k^2+p^2+q^2\right)^2\sin\left(\eta k\right)}{\eta k p^3 q^3}$$

$$\begin{aligned}[Si,\ w_1]:\ &\frac{3\cos\left(\eta k\right)}{\eta k p^3 q^3 \mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)}\left(p^4\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\right.\\
&+q^4\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\\
&+9k^4\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\\
&-6k^2p^2\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\\
&-6k^2q^2\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\\
&\left.+2p^2q^2\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\right)\end{aligned}$$

$$\begin{aligned}[Si,\ w_4]:\ &\frac{3}{2\eta k p^3 q^3 \mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)}\left(p^4\cos\left(\eta k\right)\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\right.\right.\\
&+p^4\cos\left(\eta k\right)\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+p^4\sin\left(\eta k\right)\mathrm{im}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+q^4\cos\left(\eta k\right)\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&+q^4\cos\left(\eta k\right)\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+q^4\sin\left(\eta k\right)\mathrm{im}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&-p^4\sin\left(\eta k\right)\mathrm{im}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&-q^4\sin\left(\eta k\right)\mathrm{im}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&-9k^4\sin\left(\eta k\right)\mathrm{im}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&+9k^4\cos\left(\eta k\right)\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&+9k^4\cos\left(\eta k\right)\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+9k^4\sin\left(\eta k\right)\mathrm{im}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\end{aligned}$$

$$
\begin{aligned}
&- 6k^2p^2\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&- 2p^2q^2\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+ 6k^2p^2\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&+ 6k^2q^2\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k+\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k-\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
[Si,\ w_3]:\ &\frac{-3}{2\eta k p^3 q^3 \mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)}\Bigg(p^4\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\\
&+ p^4\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+ p^4\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+ q^4\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&+ q^4\cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&+ q^4\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\\
&- p^4\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)\\
&- q^4\sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k-\frac{\sqrt{3}\eta p}{3}+\frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k+\frac{\sqrt{3}\eta p}{3}-\frac{\sqrt{3}\eta q}{3}\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 9k^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 9k^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 9k^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 9k^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 2p^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 6k^2p^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 6k^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\Big)
\end{aligned}
$$

$$
\begin{aligned}
[Si,\ w_2]:\ &\frac{-3}{2\eta k p^3 q^3 \mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)}\Big(p^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} -\right.\\
&+ p^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ p^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ q^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)
\end{aligned}
$$

$$
\begin{aligned}
&+ q^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ q^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- p^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- q^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 9k^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 9k^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 9k^4 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 9k^4 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2p^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- 6k^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&- 2p^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2 \cos(\eta k)\,\mathrm{re}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 2p^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 6k^2p^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\\
&+ 6k^2q^2 \sin(\eta k)\,\mathrm{im}\left(\mathrm{Si}\left(-\eta k + \frac{\sqrt{3}\eta p}{3} - \frac{\sqrt{3}\eta q}{3}\right)\right)\mathrm{Si}\left(\eta k - \frac{\sqrt{3}\eta p}{3} + \frac{\sqrt{3}\eta q}{3}\right)\Bigg)
\end{aligned}
$$

Elementary part ($x_a = a\eta/\sqrt{3}$, and $\sin k\eta, \cos k\eta$):

$$
\begin{aligned}
&\cos^2(k\eta)\cos(x_p)\cos(x_q)\Big\{\Big[\frac{72}{p^2q^2}\Big]\eta^{-2}\Big\}\\
&+\sin(k\eta)\Big\{\Big[-\frac{12\left(-3k^2+p^2+q^2\right)}{kp^2q^2}\Big]\eta^{-1}\Big\}\\
&+\sin^2(k\eta)\cos(x_p)\cos(x_q)\Big\{\Big[\frac{72}{p^2q^2}\Big]\eta^{-2}\Big\}\\
&+\sin^2(k\eta)\sin(x_p)\cos(x_q)\Big\{\Big[-\frac{72\sqrt{3}}{p^3q^2}\Big]\eta^{-3}\Big\}\\
&+\sin^2(k\eta)\sin(x_p)\sin(x_q)\Big\{\Big[\frac{216}{p^3q^3}\Big]\eta^{-4}\\
&\qquad+\Big[\frac{36\left(-3k^2+p^2+q^2\right)}{p^3q^3}\Big]\eta^{-2}\Big\}\\
&+\sin^2(k\eta)\sin(x_q)\cos(x_p)\Big\{\Big[-\frac{72\sqrt{3}}{p^2q^3}\Big]\eta^{-3}\Big\}\\
&+\sin(x_p)\cos^2(k\eta)\cos(x_q)\Big\{\Big[-\frac{72\sqrt{3}}{p^3q^2}\Big]\eta^{-3}\Big\}\\
&+\sin(x_p)\sin(x_q)\cos^2(k\eta)\Big\{\Big[\frac{216}{p^3q^3}\Big]\eta^{-4}\\
&\qquad+\Big[\frac{36\left(-3k^2+p^2+q^2\right)}{p^3q^3}\Big]\eta^{-2}\Big\}\\
&+\sin(x_q)\cos^2(k\eta)\cos(x_p)\Big\{\Big[-\frac{72\sqrt{3}}{p^2q^3}\Big]\eta^{-3}\Big\}
\end{aligned}
$$